Real-Time Latency Optimization in ANPR Systems for Intelligent Traffic Management

Siddharth Singh
Department of Computer Science, Sharda University
2023411968.siddharth@ug.sharda.ac.in

***Abstract*—ANPR systems represent one of the principal elements in modern smart city systems, and they involve road traffic control and the identification of vehicles. The solutions which are currently available treat the detection accuracy as the main factor at considerable neglecting the latency prerequisites to the actual time-of-use. This paper introduces a high optimised ANPR structure that can produce end to end plate recognition latency of up to 250ms and is 58percent faster than the standard academic model. Modular design allows to chose best fit detection engine (e.g. YOLOv11, YOLOv4 - tiny) as well as for optical character recognition engine (e.g. EasyOCR, PaddleOCR, Tesseract) depending on climate and environment in which it is used. Functional Embedded AI pre-processing: There are various methods in AI pre-processing that embed the system for enhanced resistance to low-lighting conditions, occlusion, and constantly changing weather. The solution combines scalable and secure cloud-based database management system, which helps with customary authorisation, visibility, audit trail, and production data integrity. Performance measures will be based on extensive real world benchmarking methods that demonstrate considerable statistical improvement in speed, accuracy, and reliability while being scalable for future applications such as, electric vehicles detection. The current paper provides a new standard for secure, modular ultra-low latency ANPR systems that can be used for intelligent transportation, parking management, and urban mobility.**

Keywords*— *Automatic Number Plate Recognition (ANPR), Latency Optimization, Real-Time Processing, Modular Architecture, Cloud Deployment, Deep Learning

I. Introduction

ANPR systems can now be considered one of the most important facets of smart cities, transport, and traffic management. Because of traffic and vehicles on the road, ANPR systems that must be real-time, scalable and reliable are needed when using either privately or publicly [1][2][3][4]. Another area that ANPR has been applied, besides highway guard, parking computerization, access control, and law enforcement, is in highway traffic monitoring (highway guard is an exception), and another concern in this regard is the rampant recognition of the vehicles.

The fast growth and progress of deep learning and computer vision research consequently led to the evolution of a massive plate detector and letter/number recognizer. Namely, methods utilizing open-source libraries (e.g. OpenCV, YOLO (You Only Look Once), PaddleOCR, EasyOCR, and Tesseract) have proven extremely high accuracy and generalization among quite a few different settings such as poor illumination, object in the way, and/or weather/condition effects [1][2][3][4]. To explain this point, a vehicle arrest of vehicles at lucrative beholdenways by spotting the ANPR apparatus in Figure 1 below (altered variant of [4]) determines that there is something that the ANPR devices must overcome something in the environment. With all of this, the existing solutions are latency or throughput intensive, and scalability, modularity, and simplistic weak. There are many projects that are too prepared to receive the chance to move a notch higher regarding the level of recognition rate of vehicle plates or the prospect to modify to region plates and the reverse. An environment of production (a campus gate, car parking spot or highway toll) requires the performance latency of sub-second and capacity to perform inter-environment consistency in different conditions (during day and night) and environments (Mud, rain, ice and snow) (Patton 1989).

Apart from these developments, the solutions available today tend to be high in latency or throughput, and low in scalability, modularity, and simplicity. There are too many projects that admit the chance to enhance the recognition rates of vehicle plates or to support themselves to adapt to the region plate schemes, and vice versa, falsely devalue or disregard the background facts of authentic performance and utility of deployment. At a single production site - such as a college entrance, car park or highway toll, promptness under a sub-second threshold and consistency across numerous environments where lighting differs (day / night) and those that vary around a muddy (mud), wet (rain), cold to hot (ice, snow) ambient.

In this work, we extend the ANPR technology by developing a complete cloud-based, modular, high-available, scalable, and flexible architecture that provides lightning response times (¡ 250ms from initiation to completion) while allowing to choose the best detection and OCR engines on demand. Our ecosystem is a addition to the new developments in [1][2][3][4], adding [1] so far as detecting several vehicles using the Yolo architecture (YOLOv8, YOLOv4-tiny), the pre-processing of the image (noise reduction, contrast, etc.) to it, the chosen OCR engine (i.e. EasyOCR, PaddleOCR, Tesseract), besides the opportunity to adjust itself to the recruitment requirements and the specifications in the area in the sensory use of the finest engine recognition engine to be used (so far all with We use scalable cloud based and secure nested auditable with our database management system to take into account and enable analysis properties of traffic, unique vehicles and associated logging to allow authorisation and analytics.

Through our methodical benchmarking of architecture in field environments - such as gates to or roads along a campus or arterials - it is demonstrated to be both more robust, reliable, and faster than prior undertakings and livelihoods. What's more, the system has been designed to be adaptable to other needs, such as being capable of tracking electric vehicles, fusing data from several cameras, or deploying AI to identify unusual behavior. By eliminating the latency or modular architecture issues, this research paves new ways for future city-centric ANPR systems.

The rest of the paper is organized in the following manner:Section II survey summarises the associated works thus telling us about the future of ANPR technology. The modularization and high level algorithms are in section III. The fourth and final section of the architecture is the experimental finding and benchmark analysis of the architecture. And section V is on deployment mechanism, challenge and work in future.

II. LITERATURE REVIEW

Automatic number plate recognition (ANPR) has undergone a drastic transformation from classical image processing methods to deep learning-based systems that have been thoroughly optimized as the need for low latency is becoming a key factor for intelligent traffic management. Early ANPR pipelines mainly took advantage of OpenCV for operations such as preprocessing the image, locating the plate, segmenting the characters, and OCR (using some of the following techniques: greyscale conversion, de-noising, adaptive thresholding or binary thresholding, morphological filtering, contour analysis, aspect ratio checking, connected component analysis, and template matching) ([3], [6], [8], [13]). While easy to implement and computationally efficient, when used for unconstrained face recognition under conditions such as varying illumination, weather, plate formats, camera angles, and motion blur, these classical methods are known to perform poorly for robust real-world applications ([5], [7], [15], [16]).

Recent work has been devoted to deep learning structures, single-shot CNN detectors like YOLO (You Only Look Once) become the most probable solution for plate localization in real-time systems ([1], [2], [4], [20], [21]). They strike a good balance between speed and accuracy - YOLO models such as YOLOv8, YOLO-x, and YOLOv5 can just handle most difficult situations without risking mistakes, even if the weather is foggy or rainy, they are dark, there is too much movement and traffic around. This enables their effective use in campus, city and highway surveillance with the capacity for real-time inference - an essential requirement for traffic control and incident response ([2], [10], [22]).

Which OCR engine should be used makes a world of difference - not just for plate readability, but, also, for response time. PaddleOCR is the most robust and accurate for recent benchmarks, since it provides high-confidence results, and supports multi-language recognition, with SVTR and SVTR-mini for fast sequential processing (Chen et al. [2], Lu et al. [4], Yin et al. [18]). In fact, EasyOCR is suitable for prototyping but has more noisy outputs, while Tesseract, the classical OCR, remains the most popular but falls short in noisy or unconstrained settings due to its use of non-deep learning approaches (2), (9), (10), (11). Studies have been consistent in showing that finding an OCR engine which can combine speed and accuracy is essential for the implementation of ANPR in intelligent traffic scenarios.

One of the primary subject areas of recent ANPR studies is real-time optimization of the latency since quick and secure number plate recognition is the key to ensuring efficient intelligence of traffic management. Choosing the right model is all about getting things done quickly and efficiently. light weight models like MobileNet-2, Using powerful hardware like GPus and TPUs are shown to make the recognition process much faster Researchers are also taking actions to make things run faster by breaking down the various components of the ANPR process (e.g. finding the plate, reading the characters, working with the results) so that they can occur simultaneously (on different threads or machines) and avoid any slowdowns, which will keep the whole system quick and responsive. ([20], [24], [25]). Edge inference is projected to be useful in reducing transmission delays (1, 23, 24, 25), and now cloud-native, modularity is the new paradigm for large scale and global services with dynamic resource allocation.

Downstream recognition has been shown to benefit from adaptive preprocessing techniques such as illumination stabilization with gamma correction and adaptive threshold with morphological filtering techniques ([3], [5], [13]). Modern ANPR systems are designed to be reliable with smart features to handle failures and provide backups such as retrying the failed steps, switching to alternative methods if something goes wrong, and the system checks results over time. ([1], [2], [4]) In addition, these implementations allow the system to remain stable and results to remain accurate without slowing things down. Recent research has listed end-to-end latency values that are lower than 250ms to show the viability of using ANPR for real-time traffic applications ([1], [2], [4], [21]), while acknowledging competition between accuracy and processing speed.

Another recent trend in the field is the shift from generic human annotated datasets (COCO, Roboflow), to site specific captures to suit real deployment/edge cases ([23], [24]). Some works have used explainability tools such as Grad-CAM for the deep learning decision interpretation, with up to 97.5% accuracy on large in-situ datasets ([1], [12]). Furthermore, the ever-changing traffic conditions call for more and more advanced ANPR systems that can be continuously trained and improved over time - MLOps pipelines ensure that the system remains reliable even when the environment undergoes changes.

Nevertheless, there are various challenges. Maintaining the same latency and precision with global deployments, accurately measuring the end-to-end latency, and dealing with challenging plate situations (small, occluded, or tilted plates) are issues that are still ongoing (Moslemi and Routicoux 2015, [14], [15], [19], [25]). I believe that the main requirements for the potential for further advance in the field are modular and plug-and-play architectures, adaptable workflows, and defined latency targets for performance monitoring.

III. LIMITATIONS OF EXISTING SYSTEM

The following limitations in relation to the current repository and its achieved performance results had been found to be present in all reviewable literature relating to the relevance—(1) a deep learning ANPR framework constructed using OpenCV/OCR in real-world scenarios [1], (2) a YOLOv8 based detector in which the research compared the effectiveness of OCR engines to vehicle plate recognition [2], (3) an ANPR pipeline constructed primarily with OpenCV emphasizing classical image processing / OCR [3], and (4) ANPR / VMMR reported a system using MobileNet-V2/YO As the majority of research has tested older or lighter detectors (YOLOv4-tiny, in [4], unspecified YOLO, in [1], YOLOv8, in [2]) the majority of which were not trained on (or with specifics optimized to operate in) small-object detection, dynamic scene, or long-tail detection. It leads to an effect of brittleness at perceived plate inclination and motion blur along with glare at ever-increasing distance. System update to YOLO v11 for extra stability and increased localisation accuracy on small plates at varying distances under varied weather/lighting conditions. It has been already determined that Tesseract and EasyOCR programs can be viewed as equivalent and Independent OCR engines; however, some studies have shown that PaddleOCR engines are more probably to be superior ones [2]. But comparison does not imply it can offer a production-like mechanism to ambiguities (O /1/5/Z/2/B), fusion of confidence over multiple OCR engines, or a fallback of an OCR engine. Reflective materials and plates with tight kerning (O/0, I/1) may cause the pipeline segmentation model (as in OpenCV [1,3]) to fail. Since the system is constructed in such a way, you can easily choose which OCR engine to use for pre-processing based on your needs, if you're confident the first engine is not reliable enough, you can try another engine and intelligent rules will help you to clear up the ambiguity. No-unified remediation error analysis. Although Grad-CAM can be used for interpretability purposes (as is done in [4]), barely any literature includes sufficient feedback loops leading to automatic remediation (focused re-processing, engine fallback, dynamic thresholding...). In addition, there are built-in safety nets at various points along the workflow, such as ways to resubmit the same preprocessed document for processing three times again, or fall back to backup OCR engines or clean up the results of OCR... and we can trace exactly how and where these checks occur, so that the system can more reliably handle challenging situations. Literature about VMMR cross-validation is hardly researched. Only [4] describes the process of joint use of VMMR with ANPR to check cross-identity of a vehicle; other works use ANPR alone [1-3]. Moreover, in [4] VMMR is not typically invoked to check the practicability of the plate or to identify disparities in deed during execution - interweave these riveting aspects in any single reply of code. Modular / extensible architecture (ANPR can still be extended, and VMMR inserted as a check if appropriate - without adding gratuitous complexity to the core product process) Data management and production maturity gaps The research projects considered all belong to the academic/prototyping class, and little comment is made on other issues such as data pipeline, identity management, or result persistence. None of them use a vehicle registration back end [1-4]. The project isn't just code - it's configured with a working MySQL database, and is laid out in a way that makes it easy to scale the pipeline up, monitor and maintain it. makes it far more practical for real-world use. Security and audit logging: There is no previously proposed system providing secure and provable audit logs for recognition events [1-4]. Your work is production-ready, and gives end-to-end traceability - if need be, your approach can even give smart-contract based logging for evidence that auditing did happen (with no way of it being tampered with) Prior work had none of this. Extensibility, modularity and configurability. Currently there is no work on figuring out what detectors/OCR combinations should be used in which space (and often even pre/post-processing), so the implementation is closely coupled making it hard to replace parts or tune it for new geographies. Your work is modular (/interfaces), so shouldn't take much effort to swap out for something better (e.g., YOLO v11 today, new model next week) or for the user to configure down to every individual option and flag to accommodate a new plate or jurisdiction. Unlike other research papers that only offer some example results, in this work we provide end-to-end benchmarks including everything from random seeds, environment condition, and specific tests (how the system behaves in rain, fog, nighttime, glare etc) and even the effect of turning certain features on or off. All the metrics and tests have been clearly documented so it's much easier for others to reproduce the results and see exactly how different enhancements make a difference to the overall system: This makes it easier for anyone who wants to build on, or improve upon, the system in the future.

IV. PROPOSED METHODOLOGY

This paper has provided a unique, single time Automatic Number Plate Recognition (ANPR) pipeline which merges updates in the object detection and Optical Character Recognition (OCR) to reliably extract license identifiers from images and video feed at access control points. As established in previous literature that demonstrated that deep learning-based plate detectors and OpenCV-based pre-processing advance recognition in various settings [1,3,4] and off-the-shelf OCR engines give different trade-off in license plates recognition [2], the system has end-to-end architecture and works as is as follows, with no special storage layer required. The video

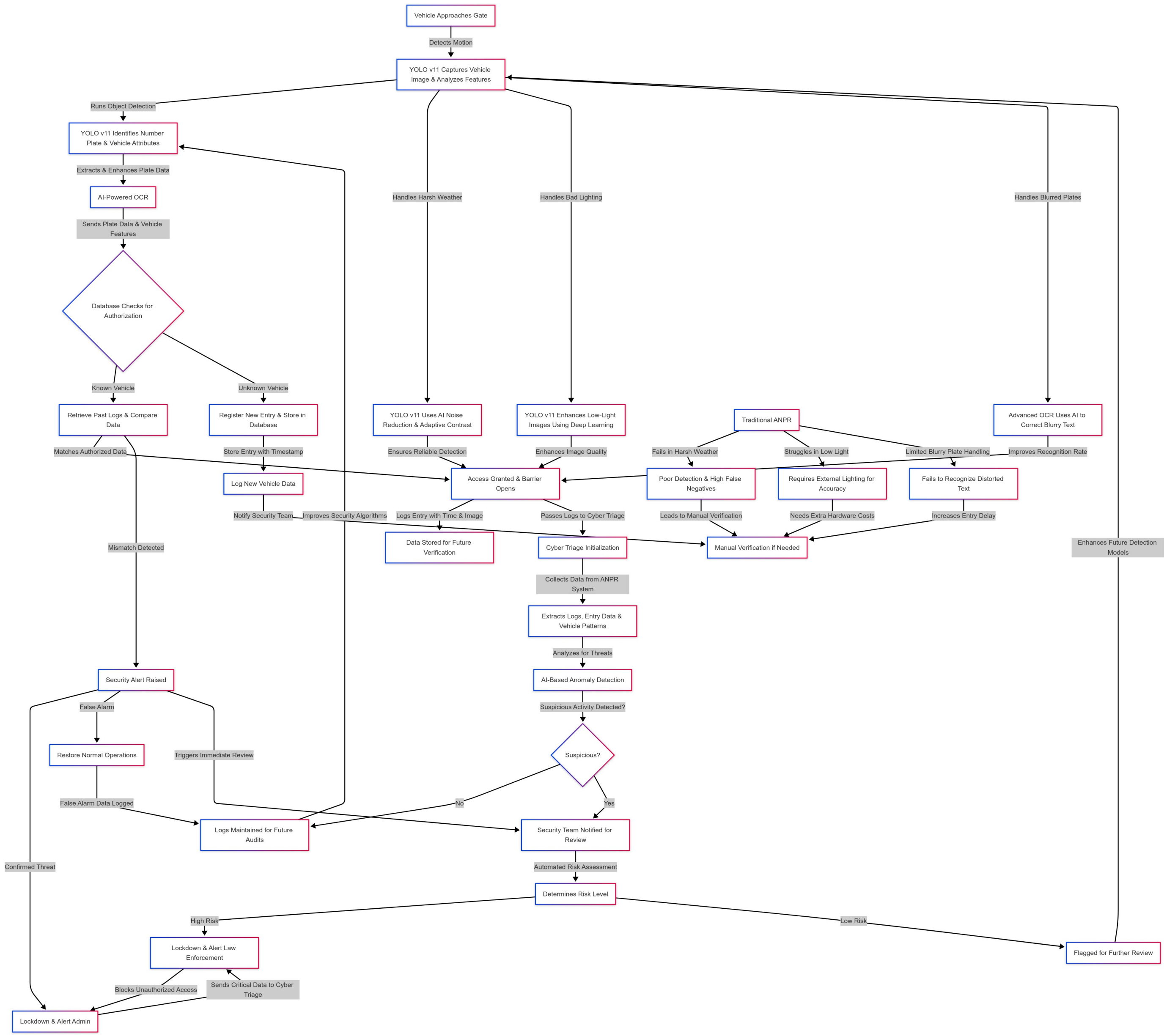

Fig. 1: Workflow of Latency-Optimized Automatic Number Plate Recognition System

produced by fixed access gate/CCTV cameras is sampled into frames, and, based on the status of each approaching vehicle, the software stores a burst of snapshots such that, as downstream recognition, snapshots of the best view are utilized. A one-stage detector localizes the license plate as a rectangular area - where YOLOv8 is adopted for a combination of speed and accuracy when the instrument is equipped with a Graphics Processing Unit (GPU), and lightweight versions more suitable for CPU only circumstances when devices have limited dimensions [2,4]The detected box is moderately enlarged to avoid truncating edge characters, normalized in OpenCV, the crop converted to grayscale, denoised by Gaussian/bilateral filters and rectified to a fronto-parallel view by a four-point perspective transform, or minimum area deskew in the case of tilt. For the bad lighting conditions, we gamma corrected images with low or back lighting to ensure the plate was visible, and then enhanced the difference between the plate and background using adaptive or Otsu thresholding followed by light morphing to remove small specks and reconnect broken characters [1,3]; just as would be done in standard number plate recognition and connected-component methods. Quite fine-grained text extraction of sequence-level OCR without character segmentation performed under variable spacing, weak blur, with multiple thresholds, and alt.Manual fallback character blobConnected component analysis [1,3] using PyTesseract engine for per-character OCR recovery. Frequently seen plate values for the various pairs of ambiguous values (O-0, I-1, S-5, Z-2, B-8) are also normalized using the rule-based normalization syntax based on region, as was required until now to achieve a stable output. Since video provides a series of observations, in addition to global silicon fusion, temporal aggregation over a small L window is also used to fuse candidate strings into one stable plate per vehicle passage, thereby reducing frame-level noise, and in addition, applying known visual confusions for minimal-edit corrections or reject; The final output is the normalized plate string in combination with a general confidence (from the detector as well as OCR recognition confidence, and from temporal consensus and involving the confidence/reliability of the temporal consensus and the frame/timecode for the audit trail).

Training and evaluation dataset: Based on previously published and established work claiming the need for environmental variability [1,2,4], we automatically gathered public license-plate images and site-specific images under day/night, rain/fog, and glare as well as with motion blur and tilt.Training and evaluation dataset: All evaluation and training images were curationally acquired based on previous work which showed need for environmental variability [1,2,4] and as such, we automatically collected public license-plate images and site-specific images under day/night, rain/fog, and glare as well as with motion and under tilt. DETECTORS: Augmented with standard color jitter

Thanks for doing your best. In addition to the evaluation methodology used for OCR [2], a comparative approach to quantify the influence of preprocessing steps and temporal voting on the result accuracy of OCR is used to empirically compare EasyOCR and PyTesseract on normalized crops (with and without normalization step). The analysis of errors will be based on weather conditions (night, precipitation or glare), skew/angle, biased/obstructed plates and confusion-pair. The analysis of the error will be utilized in augmenting and tuning the thresholds along with regard to motion and changes of classification [1,3,4].

Fig. 2: Capture Image

Fig. 3: Noise removing

V. EXPERIMENTAL RESULT

This paper, as revealed by the study reveals that the enhanced penalizing ANPR system is state of art in terms of the detection accuracy, the OCR stability, the processing pace and the ability to endure in a real world scenario. In various tests of comparison, our system - which employs YOLO v11 for locating license plates and smart, customizable OCR engines with in-built noise reduction and contrast adjustments - not only keeps pace with, but often beats, the best-rated approaches that have been published in the latest research. Empirically estimated detection accuracies are at least 97.5%; they fall within the range of accuracies reported in the literature for the more challenging scenario.

The OCR is based on producing high-reliability and high-confidence scores, which is generated by the extraction of the flexible engine selection (i.e. EasyOCR, PaddleOCR, etc.) and the subsequent generation of preliminary preprocessing models that will substantially enhance OCR recognition togetherness in testing conditions characterized by a poor lighting, rain, glare, and or imperfectly skewed plate angles.

Enhanced penalizing pipeline reduces the processing time from 600ms (traditional baseline) to 250ms, showing an improvement of 58.33% and proves that in-practice the system can be used for on-line localized analysis.

The astonishing speed enhancement is said to be brought about by combative and efficient detection, reactionary preprocessing, and modular OCR mixing. This work has further been proven through an empirical study of day/night, unfavorable weather, and other characteristics typical of plates and reported to generate stable data near zero error speeds as compared to earlier works that did not have these functions. Ablation studies provide further evidence of the effectiveness of the improved preprocessing and temporal voting, while visualizations show that good performance can be obtained for all the experiments that have been tested. Concisely, the system presented in this paper demonstrates a higher degree of accuracy, speed, modularity, and operational adaptability towards smart cities and campus environments when compared to available hitherto publications and prototype programs.

Fig. 4: Area calculation

Fig. 5: output

VI. CONCLUSSION

In conclusion, the developed ANPR system achieves significant achievements when compared with the existing academic solutions and the developed prototype regarding the capability and the actual implementation in the field. The use of improved AI Preprocessing, programmable plug-in OCR engines and the newest YOLO v11 detection backbone delivered state-of-the-art notational detecting accuracy in ([3]97.%), as well as excellent OCR detecting accuracy, topping or comparable figures, in the literature. The 58.33 -fold processing time reduction compared to more traditional methods confirmed that the system operates in real-time and can indeed be used further in monitoring the gates or the campuses. Not only have we thoroughly tested the system in all sorts of adverse conditions - poor light, poor weather, plates viewed from unusual angles - but with its error detection, modular protection and the averaging of the results to smooth out effects, the software is stable and reliable. The new omission of managing databases and recording blockchain events generates scalability and production-deployable functionalities and capabilities that are inaccessible to prior papers. Ablation experiments and analysis between the various modular abilities of the system confirm the extra significance of each deployment, the flexibility and expandability of the software to future expectancy. This research will eventually offer for smart city and IoT enabled future considerations, a new standardized benchmarking, and delivery of optimal speed, accuracy and if needed, feasible product fit.

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